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Shareholder Investment Horizons and the Market for Corporate Control

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1 Big picture

- **Question.** Does the investment horizon of a firm's institutional shareholders affect how the firm behaves in the market for corporate control? More specifically:
 - Are firms with short-term shareholders more likely to become takeover targets?
 - Do target firms with short-term shareholders receive lower acquisition premia?
 - Do bidder firms with short-term shareholders make worse acquisitions?
- **Setting.** The paper studies U.S. mergers and acquisitions from **1980 to 1999**. It combines:
 - SDC takeover announcements,
 - CRSP stock returns,
 - COMPUSTAT accounting data,
 - and CDA/Spectrum institutional ownership filings.

The final M&A event sample contains **3,814 acquisition announcements** involving U.S. public targets.

- **Core challenge.** “Investor horizon” is not directly observable. A shareholder can be a pension fund, mutual fund, investment adviser, or activist institution, but its legal type is not the same thing as its actual willingness to hold shares and monitor managers. The paper therefore constructs a horizon measure from the investor's **actual portfolio turnover**.
- **Main contribution.** The authors introduce an empirical measure of shareholder investment horizon based on institutional investors' **portfolio churn rates**. They then use this measure to show that shareholder horizons matter for both sides of the takeover market:
 - for targets, short-term shareholders reduce bargaining power and takeover premia;
 - for bidders, short-term shareholders weaken monitoring and allow more value-destroying acquisitions.
- **Main mechanism.** The paper emphasizes two related channels:
 1. **Bargaining power.** Long-term target shareholders can hold out for a higher price. Short-term shareholders are more willing to sell quickly, so the target has a weaker bargaining position.
 2. **Monitoring.** Long-term shareholders have stronger incentives to monitor managers because they expect to remain invested long enough to capture the benefits. Short-term shareholders are more likely to exit instead of monitor.
- **Why it matters.** The paper speaks to the debate on **short-termism** in U.S. corporate governance. If short-term shareholders weaken monitoring and bargaining, then the identity and horizon of shareholders affect real corporate decisions, not only stock trading.
- **Main empirical question.** Do differences in shareholder horizons explain:
 - the premium target shareholders receive,
 - the announcement return of bidders,
 - the likelihood that a firm becomes a target or bidder,

- and the long-run post-merger performance of acquirers?
- **Main empirical results.**
 - Target firms with short-term shareholders are **more likely to receive a takeover bid**.
 - Conditional on receiving a bid, target firms with short-term shareholders receive **lower acquisition premia**.
 - A one-standard-deviation increase in target shareholder turnover reduces the target premium by roughly **3 percentage points**; this is equivalent to reducing investors' average holding period from about **15 months to 11 months**.
 - Bidder firms with short-term shareholders experience **more negative abnormal returns** around merger announcements.
 - Acquirers with short-term shareholders also show stronger **long-run underperformance**; over three years, a value-weighted strategy that buys high-turnover acquirers and sells low-turnover acquirers earns about **−0.7% per month**, or roughly **−8% per year**.
 - The target-premium result survives controls for liquidity, ownership concentration, investor type, arbitrage capital, managerial holdings, and Heckman sample-selection correction.
- **Economic takeaway.** Short-term shareholders make target firms easier and cheaper to buy, but they also make bidder firms more vulnerable to managerial overbidding. Long-term capital is valuable because it strengthens both **bargaining** and **monitoring**.

2 Data and measurement (key objects)

2.1 M&A sample

- **Source of deal data.** Acquisition announcements come from the **Securities Data Corporation (SDC)** M&A database.
- **Sample period.** The paper covers deals announced between **January 1980 and December 1999**.
- **Target requirements.**
 - The target must be a U.S. firm listed on the **NYSE, Amex, or Nasdaq**.
 - The target must have a CUSIP that can be matched to CRSP.
 - The deal outcome must be known: completed or withdrawn.
 - The transaction value must be at least **1%** of the target's market value.
- **Multiple bids.** When several bids occur for the same target within one year of the first bid, the paper keeps only the **first bid**. This prevents revised or competing bids from mechanically contaminating the premium measure.
- **Final event sample.** The baseline sample contains **3,814 events**.
- **Deal characteristics.** The paper records whether a deal is completed, hostile, all-cash, all-stock, intra-industry, a tender offer, and whether the bidder already owns a toehold in the target.

2.2 Stock return, accounting, and institutional ownership data

- **CRSP.** Used for stock prices, returns, shares outstanding, market values, and abnormal returns.
- **COMPUSTAT.** Used for firm-level accounting controls such as return on equity, sales growth, liquidity, leverage, market-to-book, price-earnings ratio, and size.
- **CDA/Spectrum 13-F filings.** Used to measure institutional ownership. The database includes quarterly holdings of institutions with more than **US\$100 million** under discretionary management.
- **Institutional holdings threshold.** Reportable positions are holdings of more than **10,000 shares** or more than **US\$200,000** in value.
- **Important limitation.** The 13-F data report long positions. They do not report short positions, which matters because merger arbitrageurs often use long-short strategies around M&A events.

2.3 Premium and return variables

- **Abnormal return premium.** The main target premium measure is the cumulative abnormal return to the target from trading days $[-63, +126]$ around the announcement date.
- **Run-up and markup.**
 - The **run-up period** is $[-63, -1]$.
 - The **markup period** is $[0, +126]$.
 - The abnormal return premium is run-up plus markup.
- **Benchmark model.** Abnormal returns are computed relative to a CRSP value-weighted market model, estimated using daily returns for the prior trading year ending on day -64 .
- **Actual offer premium.** The alternative premium measure is

$$\frac{\text{bidder's offer}}{\text{target's pre-bid market value of equity}} - 1.$$

The bidder's offer is constructed from SDC consideration and offer-price fields.

- **Bidder abnormal return.** The same abnormal-return logic is applied to bidders. The paper studies both an extended Schwert-style window and a short $[-1, +1]$ announcement window.

2.4 Investor horizon and other shareholder variables

- **Main variable: investor turnover.** A firm has short-term shareholders if its institutional owners tend to rotate their overall portfolios frequently.
- **Fraction.** The fraction of the firm's shares held by institutional investors.
- **Ownership concentration.** A Herfindahl index over the distribution of institutional shareholders' stakes in the firm.
- **Manager concentration.** The weighted average concentration of each institutional investor's own portfolio. This proxies for exposure to idiosyncratic risk.

- **Industry exposure.** The weighted average fraction of each institutional investor's portfolio invested in the event firm's industry. This proxies for industry-specific information and monitoring ability.
- **Beta.** The weighted average beta of institutional investors' portfolios.
- **Activism.** The fraction of firm shares held by activist institutions, defined using public pension funds that are members of the Council of Institutional Investors.
- **Arbitrage capital.** A measure of capital supplied by merger arbitrageurs, following the idea that arbitrageurs may enter target stocks after takeover announcements.

2.5 Timing of measurement

- **Predetermined horizon measure.** Shareholder variables are measured so that at least **two full quarters** pass between the horizon measurement and the acquisition announcement.
- **Practical timing.** Investor turnover is measured **six to nine months before** the announcement date.
- **One-year history.** The firm-level turnover measure averages investor churn rates over four quarters, so it uses roughly one year of portfolio behavior.
- **Why this matters.** Because churn rates are changes in holdings, the measure uses information reaching back about **18–21 months** before the event. This makes the horizon variable less likely to be mechanically driven by rumors or trading in anticipation of the specific deal.

2.6 Key descriptive facts

- In the baseline sample, **84.6%** of deals are successful and **6.1%** are hostile.
- **63.2%** of deals are all-cash, **15.0%** are all-stock, and **31.4%** are intra-industry deals.
- The average abnormal return premium is **21.5%**; the average actual offer premium is **52.7%**.
- The median relative size of target to bidder is **19.5%**; the median number of days to completion is **59**.
- Median investor turnover is similar for targets and bidders: about **0.37** for targets and **0.39** for bidders.
- Median institutional ownership is much higher for bidders than for targets: about **49%** for bidders versus **30%** for targets.
- A median investor turnover rate around **0.39** means institutional investors churn roughly **80%** of a position per year, corresponding to an average holding period of about **15 months**.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic hypothesis: why horizons should matter

- **Null hypothesis.** If capital markets are perfect and managers always maximize shareholder value, then shareholder horizons should not matter. Shareholders can simply rebalance portfolios after the merger.
- **Alternative hypothesis.** Shareholder horizons matter because takeovers involve bargaining, monitoring, and managerial agency problems.
- **Target-side prediction.** If target shareholders are short term:
 - they have less ability or willingness to hold out for the full value of the target;
 - target managers face weaker monitoring;
 - bidders expect lower resistance;
 - the target is more likely to receive a bid but receives a lower premium.
- **Bidder-side prediction.** If bidder shareholders are short term:
 - bidder managers face weaker monitoring;
 - managers have more discretion to pursue empire building or overbid;
 - the bidder is more likely to make a bid;
 - bidder announcement returns and long-run returns are worse.
- **Cross-effect prediction.** The payoff to one side of the transaction should depend on the shareholder horizon of the other side. Short-term bidder shareholders should benefit the target, while short-term target shareholders should benefit the bidder.

3.2 Institutional investor churn rate

For institutional investor i , let Q_i be the set of stocks in its portfolio, $N_{j,i,t}$ the number of shares of firm j held by investor i in quarter t , and $P_{j,t}$ the price of stock j . The quarterly churn rate is

$$CR_{i,t} = \frac{\sum_{j \in Q_i} |N_{j,i,t}P_{j,t} - N_{j,i,t-1}P_{j,t-1} - N_{j,i,t-1}\Delta P_{j,t}|}{\sum_{j \in Q_i} \frac{N_{j,i,t}P_{j,t} + N_{j,i,t-1}P_{j,t-1}}{2}}.$$

Interpretation.

- The numerator measures active changes in the investor's positions, adjusted for passive changes in value due to price movements.
- The denominator scales trading by average portfolio size.
- A higher $CR_{i,t}$ means the investor rotates its portfolio more quickly and has a shorter effective investment horizon.

3.3 Firm-level investor turnover

Let S_k be the set of institutional shareholders in firm k , and let $w_{k,i,t}$ be investor i 's weight in total institutional ownership of firm k . The firm-level investor turnover measure is

$$IT_k = \sum_{i \in S_k} w_{k,i,t} \left(\frac{1}{4} \sum_{r=1}^4 CR_{i,t-r+1} \right).$$

Interpretation.

- This is a weighted average of the portfolio churn rates of the firm's institutional shareholders.
- The weights are based on each investor's importance in the firm's institutional shareholder base.
- A firm has short-horizon shareholders when its large institutional shareholders churn their overall portfolios frequently.
- The measure is not the trading volume of the event stock itself; it is based on the investors' behavior across their whole portfolios.

3.4 Target premium regression

The baseline cross-sectional specification for target firms is

$$Premium_k = X_k^{IT} \beta + X_k^{SH} \delta + X_k^C \gamma + u_k.$$

- $Premium_k$: target abnormal return premium or actual offer premium.
- X_k^{IT} : investor turnover of the target, sometimes also including bidder turnover as a cross-effect.
- X_k^{SH} : other shareholder variables, including institutional ownership, concentration, activism, investor beta, industry exposure, and arbitrage capital.
- X_k^C : standard M&A controls such as size, market-to-book, leverage, liquidity, hostility, competing bids, tender offer, intra-industry status, and toehold.

Interpretation.

- The coefficient on target investor turnover is the central estimate.
- A negative coefficient means that short-term target shareholders are associated with lower takeover premia.
- This supports the bargaining/monitoring channel: short-horizon shareholders make the target easier to acquire cheaply.

3.5 Bidder announcement-return regression

The same specification is applied to bidder firms:

$$CAR_k^{Bidder} = X_k^{IT} \beta + X_k^{SH} \delta + X_k^C \gamma + u_k.$$

Interpretation.

- The dependent variable is the bidder's abnormal stock return around the acquisition announcement.
- A negative coefficient on bidder investor turnover means that bidders with short-term shareholders experience worse market reactions.
- This is consistent with weak monitoring: short-term shareholders allow managers to overpay or pursue acquisitions that do not maximize bidder shareholder value.

3.6 Acquisition likelihood model

The paper also estimates whether investor turnover predicts the likelihood that a firm becomes a target or bidder. Let $h_k = 1$ if firm k enters a deal and $h_k = 0$ otherwise. The latent variable model is

$$h_k^* = X_k^{IT} \theta + X_k^{SH} \eta + X_k^C B + u_k, \quad h_k = 1\{h_k^* > 0\}.$$

Interpretation.

- For targets, a positive investor-turnover coefficient means short-term shareholder bases make firms more likely to receive bids.
- For bidders, a positive coefficient means short-term shareholder bases make firms more likely to initiate acquisitions.
- This likelihood equation is also used as the first stage in the sample-selection correction.

3.7 Heckman sample-selection correction

Because investor horizons affect whether a deal is observed in the first place, the paper checks whether the premium regressions suffer from selection bias. The second-stage equation is

$$Premium_k = X_k^{IT} \beta_2 + X_k^{SH} \delta_2 + X_k^C \gamma_2 + \sigma \lambda_k + u_k,$$

where λ_k is Heckman's inverse Mills ratio constructed from the first-stage probit.

Interpretation.

- This asks whether the target-premium result survives after accounting for the fact that short-term shareholder firms are more likely to appear in the takeover sample.
- The main coefficient remains negative for target investor turnover and positive for bidder investor turnover.
- Thus the lower-premium result is not only a sample-selection artifact.

3.8 Long-run post-merger performance

The paper studies long-run acquirer returns using calendar-time portfolio methods. In the calendar-time portfolio regression approach, the excess return of a portfolio of acquirers is regressed on Fama–French factors:

$$R_{p,t} - R_{f,t} = \alpha + \beta_m(R_{m,t} - R_{f,t}) + \beta_s SMB_t + \beta_h HML_t + \varepsilon_t.$$

Interpretation.

- α measures abnormal long-run performance after controlling for standard risk factors.
- The paper compares acquirers with high investor turnover to acquirers with low investor turnover.
- A negative high-minus-low α means that acquirers with short-term shareholders underperform acquirers with long-term shareholders after the deal.

The calendar-time abnormal return approach first constructs firm-level residuals:

$$\hat{e}_{i,t} = R_{i,t} - R_{f,t} - \hat{\beta}_{m,i}(R_{m,t} - R_{f,t}) - \hat{\beta}_{s,i}SMB_t - \hat{\beta}_{h,i}HML_t.$$

Interpretation.

- These residuals are aggregated into monthly portfolios.
- The method is designed to reduce problems caused by event clustering and cross-sectional dependence.
- The main long-run conclusion is the same: high-turnover acquirers underperform.

4 Identification and instruments (what makes the estimates credible)

4.1 What the paper is trying to identify

- The paper wants to identify whether **shareholder horizon**, not just institutional ownership in general, matters for M&A outcomes.
- The key object is the relation between a firm's pre-deal investor turnover and subsequent takeover outcomes:
 - bid likelihood,
 - target premium,
 - bidder announcement return,
 - and bidder long-run performance.
- The empirical design is not a randomized experiment. The credibility comes from careful timing, rich controls, cross-effects, and selection corrections.

4.2 Predetermined measurement of investor horizon

- The investor-turnover variable is measured at least two full quarters before the announcement.
- It averages four quarters of the investors' total portfolio behavior, not just trading in the event firm.
- This timing reduces the concern that the measure is mechanically driven by takeover rumors, run-up trading, or arbitrage activity immediately before the announcement.
- The authors also report tests showing no economically meaningful pre-announcement positioning by institutional investors in event firms.

4.3 Why portfolio-level churn is useful

- A firm's trading volume could reflect pending news about the firm itself.
- By contrast, an institutional investor's total portfolio churn is closer to an investor-level style or mandate.
- The firm-level investor turnover variable then summarizes the horizon of the firm's shareholder base, rather than event-specific trading intensity.

4.4 Controls for alternative explanations

- **Institutional ownership level.** The paper controls for the fraction of shares held by institutions, because ownership level itself can affect monitoring.
- **Ownership concentration.** It controls for the Herfindahl concentration of institutional ownership, because large blockholders may monitor more effectively.
- **Investor risk preferences.** It controls for investor portfolio beta and portfolio concentration.
- **Information and expertise.** It controls for investors' exposure to the event firm's industry.
- **Investor type.** It checks whether the result is simply driven by mutual funds or investment advisers.
- **Activism.** It controls for stakes held by activist institutional investors.
- **Arbitrage.** It controls for changes in arbitrage capital and checks that results survive excluding merger arbitrageurs from the turnover measure.
- **Managerial ownership.** It adds managerial holdings in robustness tests, where data are available.
- **Standard M&A controls.** It includes size, market-to-book, leverage, liquidity, profitability, hostility, competing bids, intra-industry status, tender offer, and toehold.

4.5 Cross-effects strengthen the interpretation

- The paper does not only ask whether target shareholder turnover lowers the target premium.

- It also tests whether bidder shareholder turnover raises the target premium.
- The positive cross-effect is important: if short-term bidder shareholders weaken bidder monitoring, bidders may overpay, which should show up as higher gains for targets.
- This pattern is consistent with a bargaining-surplus interpretation rather than a purely mechanical relation between target characteristics and premia.

4.6 Sample-selection correction

- A key concern is that short-term target shareholders may make a firm more likely to receive a bid, so the observed takeover sample is selected.
- The paper estimates acquisition likelihood on a large firm-year universe created by merging CRSP, COMPUSTAT, and CDA/Spectrum.
- It then uses Heckman's correction to estimate premium regressions conditional on selection into the M&A sample.
- The core coefficients survive, so the main premium results do not appear to be driven by selection alone.

4.7 Interpretation and remaining limitations

- The evidence is best interpreted as strong empirical support for a horizon-monitoring-bargaining channel.
- However, shareholder horizon is not randomly assigned. Firms with short-term investors may differ in unobserved ways from firms with long-term investors.
- The paper addresses this by measuring horizons before the event, controlling for many firm and shareholder characteristics, using cross-effects, checking likelihood and selection, and studying long-run outcomes.
- The results therefore show that shareholder horizon is an economically important state variable in corporate control, even if the design is not a clean natural experiment.

5 Tables and figures

5.1 Table 1: summary statistics for the M&A sample

Read:

- The base sample has **3,814** M&A events from 1980–1999.
- The 1990s have many more events than the 1980s: **2,582** versus **1,232**.
- Deal hostility declines sharply from the 1980s to the 1990s: **10.8%** hostile in the 1980s versus **3.9%** in the 1990s.
- Stock-financed deals become more common in the 1990s, while all-cash deals remain common.

- The average abnormal return premium is **21.5%**, while the average actual offer premium is **52.7%**.
- The table establishes that the sample resembles the standard U.S. M&A samples used in the takeover literature.

Table 1

Summary statistics for the sample used in this study

Our base sample consists of 3,814 events recorded in the SDC mergers and acquisitions database from 1980 to 1999 that have non-missing data for the variables used in the regressions in the following tables. We keep acquisition announcements that are the first bid for a target in each contest. For each company involved in the event, we obtain the corresponding accounting variables (from COMPUSTAT), stock price behavior (from CRSP), and institutional investor variables (from CDA/Spectrum and our previous calculations). N denotes the number of cases (observations) for which the variable in question is present in the sample, and the numbers provided are sample averages (unless otherwise noted). An acquisition is considered successful if the "status" field in SDC has "completed" as keyword. An acquisition is considered hostile if the "attitude" field in SDC was marked "unsolicited" or "hostile". The "consideration offered" field in SDC refers to the list of all components of consideration (i.e., means of payment) offered by the acquirer/bidder. An acquisition is considered all-cash if the field "consideration offered" in SDC had only "cash" as keyword. An acquisition is considered all-equity if the field "consideration offered" in SDC did not include "cash" as keyword. To define intra-industry acquisitions, we use the Fama and French (1997) 49-industry classification. The SIC code used to classify firms was obtained from CRSP. Abnormal return premium is the premium, defined as the cumulative abnormal return, measured relative to a CRSP value-weighted market model using a year of prior daily data, to the target firm stock for trading days $[-63, +126]$ relative to the announcement date (see Schwert, 2000). Actual offer premium is the premium defined as [bidder's offer/target's pre-bid market value of equity] $- 1$, where the value of the bidder's offer is computed using, in order of availability, the sum of the value of the considerations offered, the initial offer price, or the final offer price as reported in SDC (see Officer, 2003, for details). Days to completion is the difference in calendar days between the announcement date and the date the deal is considered effective.

	N	Full sample	1980–1989	1990–1999
Number of events	—	3,814	1,232	2,582
Percent successful	3,814	84.6%	77.5%	88.0%
Percent hostile	3,814	6.1%	10.8%	3.9%
All cash	3,814	63.2%	64.2%	62.7%
All stock	3,814	15.0%	7.9%	18.4%
Intra-industry	3,814	31.4%	21.7%	36.0%
Target listed on Nasdaq	3,814	51.7%	33.0%	60.7%
Abnormal return premium	3,814	21.5%	16.3%	23.9%
Actual offer premium	2,501	52.7%	48.4%	54.4%
Acquirer leverage > target	1,453	74.3%	77.4%	73.3%
Acquirer M/B > target M/B	1,410	65.7%	59.1%	67.9%
Relative size (median)	1,532	19.5%	25.3%	17.7%
Days to completion (median)	3,227	59	49	64

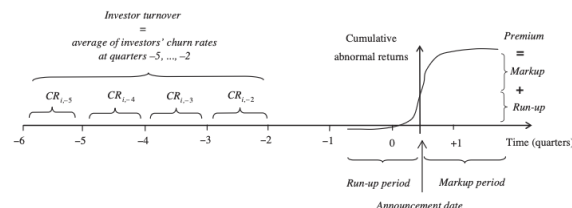


Fig. 1. Investor turnover, our main independent variable, is the average of investors' churn rates for a one-year period that finishes at least two full quarters before the announcement date. The run-up period is defined as the interval of trading days $[-63, -1]$ relative to the announcement date (Schwert, 2000). Markup is analogously defined, for trading days $[0, 126]$. Our main dependent variable, abnormal return premium, is the sum of abnormal returns during the run-up and the markup period.

5.2 Figure 1: timing of investor turnover and premium measurement

Read:

- The figure shows the timing of the main independent variable and the main dependent variable.
- Investor turnover is measured using churn rates from quarters -5 to -2 , so the measure is set before the announcement-window return measurement.
- The run-up period is $[-63, -1]$, and the markup period is $[0, +126]$.
- This figure is central because it explains why the paper treats investor turnover as predetermined relative to the acquisition announcement.

5.3 Table 2: shareholder variables for targets and bidders

Read:

- Target and bidder investor turnover are similar in magnitude: mean target turnover is **0.37**; mean bidder turnover is **0.38**.
- Bidders have much higher institutional ownership: median institutional ownership is **49%** for bidders and **30%** for targets.

- Ownership is more concentrated among targets than bidders: target median concentration is **0.14**; bidder median concentration is **0.05**.
- Activist investors hold larger median stakes in bidders than in targets.
- The key descriptive point is that the investor-turnover measure varies substantially across firms and is not simply the same as institutional ownership level or concentration.

Table 2

Summary statistics of the shareholder variables for target and bidder firms

Denote by S the set of shareholders in company k and by $w_{i,t}$ the weight of investor i in the total percentage held by institutional investors in quarter t . The instant of measurement t is such that there is at least two full quarters between the measurement of all shareholder variables and the announcement date. The investor turnover of firm k is the weighted average of the average total portfolio churn rates of its investors over four quarters (see text for details on the construction of the churn rates). Investor turnover of firm $k = \sum_{i \in S} w_{i,t} \left(\frac{1}{4} \sum_{t=1}^4 CR_{k,t} \right)$. Fraction denotes the ratio of a firm's shares held by institutional investors relative to total shares outstanding in CRSP. Concentration is the Herfindahl index calculated over the distribution of weights $w_{i,t}$. For each institutional investor, we calculate the Herfindahl index and the average beta of its portfolio. Manager concentration is the weighted average (using the weights $w_{i,t}$) of the Herfindahl index for all investors $i \in S$. Similarly, beta is the weighted average (using the weights $w_{i,t}$) of the betas of all investors $i \in S$. The activism variable is the fraction of the firm's shares held by activist institutional investors relative to total shares outstanding. Activist investors are the ones belonging to the list of members of the Council of Institutional Investors. We also include in this table a measure of the change in arbitrage capital (following Baker and Savasoglu, 2002), measured in percentage terms. Arbitrage capital is the sum of total portfolio holdings of institutions considered as arbitrageurs. An institution is considered an arbitrageur if its holdings go from zero to positive in a target firm in the quarter immediately following a takeover announcement, for more than two hundred events during the sample period. This variable takes only one value per quarter, the averages presented are therefore time-series averages during the sample period.

	N	Full sample			1980-1989			1990-1999		
		Mean	Median	Standard error	Mean	Median	Standard error	Mean	Median	Standard error
Target										
Investor turnover	3,736	0.37	0.37	0.15	0.42	0.44	0.14	0.35	0.32	0.15
Fraction	3,746	0.33	0.30	0.23	0.28	0.24	0.20	0.35	0.32	0.23
Concentration	3,746	0.21	0.14	0.20	0.33	0.15	0.23	0.30	0.13	0.19
Manager concentration	3,746	0.02	0.01	0.03	0.02	0.02	0.03	0.02	0.01	0.03
Industry exposure	3,746	0.05	0.04	0.05	0.04	0.03	0.04	0.05	0.04	0.05
Beta	3,746	1.13	1.10	0.15	1.08	1.07	0.10	1.13	1.13	0.16
Activism	2,864	0.05	0.03	0.07	0.05	0.03	0.05	0.05	0.03	0.08
Bidder										
Investor turnover	2,162	0.38	0.39	0.14	0.43	0.45	0.14	0.36	0.32	0.13
Fraction	2,168	0.47	0.49	0.24	0.37	0.39	0.21	0.52	0.55	0.24
Concentration	2,168	0.11	0.05	0.16	0.13	0.06	0.19	0.10	0.05	0.14
Manager concentration	2,168	0.02	0.01	0.02	0.02	0.02	0.02	0.02	0.01	0.03
Industry exposure	2,168	0.06	0.05	0.05	0.05	0.04	0.05	0.06	0.05	0.05
Beta	2,168	1.12	1.10	0.13	1.07	1.07	0.09	1.15	1.11	0.14
Activism	1,987	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06	0.06
Change in arbitrage capital	3,184	0.05	0.06	0.09	0.04	0.06	0.10	0.06	0.06	0.08

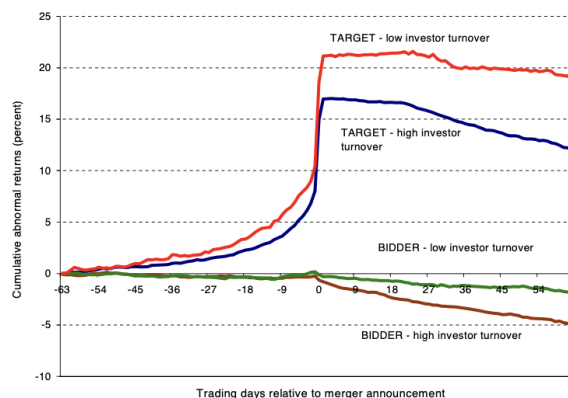


Fig. 2. Cumulative abnormal returns around acquisition announcement: firms with high investor turnover versus firms with low investor turnover. Cumulative abnormal returns for target and bidder firms are measured relative to a CRSP value-weighted market model using a year of prior daily data (Schwert, 2000). Investor turnover is the average of investors' churn rates for a one-year period that finishes at least two full quarters before the announcement date. Each month we sort all stocks in the CRSP-COMPUSTAT-CDA/Spectrum universe according to their investor turnover. A target is considered a high investor turnover firm if its investor turnover variable is in the top third of the distribution for the entire universe on the month prior to the acquisition. Inversely, a target is considered a low investor turnover firm if its investor turnover variable is in the bottom third of the distribution for the entire universe in the month prior to the acquisition. The same procedure is used for the case of bidder firms.

5.4 Figure 2: abnormal returns by high and low investor turnover

Read:

- The paper sorts firms into high-turnover and low-turnover shareholder bases.
- For targets, high-turnover shareholder bases are associated with lower cumulative abnormal returns around the acquisition.
- For bidders, high-turnover shareholder bases are associated with more negative abnormal returns.
- The figure gives visual evidence for the main regression results: short-term shareholders weaken target bargaining and bidder monitoring.

5.5 Table 3: target premium regressions

Read:

- The main coefficient is the coefficient on target investor turnover.
- In the baseline specification, the coefficient is about -0.20 with a t -statistic of about -2.2 .
- A one-standard-deviation increase in target investor turnover reduces the target premium by about **3 percentage points**.

Dependent variable	Abnormal return premium												Actual offer premium			
	(1)				(2)				(3)				(4)			
	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic
Intercept	0.905	11.92***	1.387	6.15***	1.228	5.18***	1.848	2.88***	1.161	3.55***	0.780	5.30***	0.780	2.13**		
ROE	0.002	0.43	0.001	0.15	-0.001	-0.13	0.027	0.36	0.001	0.13	0.002	0.44	-0.007	-1.49		
Sales growth	-0.024	-1.10	-0.073	-2.48***	-0.042	-1.15	-0.125	-1.29	-0.029	-0.64	-0.017	-0.71	-0.042	-1.29		
Liquidity	-0.006	-0.16	-0.008	-0.15	-0.035	-0.81	-0.055	-0.35	-0.077	-1.09	-0.020	-0.42	-0.182	-2.65***		
D/E	0.022	0.44	0.132	2.20**	0.077	1.34	-0.217	-1.50	0.068	0.94	0.090	1.64	0.062	0.88		
M/B	-0.006	-2.27**	-0.006	-1.18*	-0.007	-2.64**	0.001	0.23	-0.007	-1.45	0.002	0.79	0.004	0.93		
P/E	0.000	-1.11	0.000	-1.14	0.000	-0.58	-0.001	-0.79	0.000	-1.04	0.000	-1.02	0.000	-0.89		
Size	-0.002	-10.62***	-0.008	-5.28***	-0.079	-8.58***	-0.086	-2.60**	-0.078	-4.68***	-0.047	-5.57***	-0.023	-2.18**		
Hostility	0.046	1.69*	0.081	2.96***	0.091	3.18***	0.105	1.35	0.034	0.90	0.055	1.77*	0.041	0.97		
Competing bids	0.072	1.71*	0.091	2.06**	0.081	1.90**	0.264	1.40	0.006	0.14	0.006	0.14	-0.006	-0.13		
Intra-industry	0.098	5.55***	0.075	3.61***	0.061	3.00***	0.020	0.33	-0.008	-0.20	0.082	4.10***	0.052	1.95*		
Tender offer	0.240	12.58***	0.207	10.06***	0.205	9.50***	0.170	1.33	0.160	1.77***	0.137	6.20***	0.043	1.44		
Threshold	-0.148	-8.04***	-0.122	-6.01***	-0.116	-5.50***	-0.147	-2.58	-0.091	-3.20***	-0.129	-5.42***	-0.163	-5.14***		
Investor turnover	-0.200	-2.20**	-0.348	-3.50***	-1.088	-3.32***	-0.560	-4.51***	-0.176	-2.00**	-0.239	-1.76*				
Fraction	0.003	1.66*	0.243	2.21**	0.040	0.16	0.273	1.95*	0.076	1.49	0.232	1.99**				
Concentration	-0.440	-1.35	-0.052	-0.34	-0.369	-0.57	0.004	0.02	0.234	2.09**	0.522	3.07***				
Manager	0.340	1.45	0.059	0.17	3.813	3.57***	0.069	0.17	-0.071	-0.15	-0.756	-1.59				
Industry exposure	0.114	0.30	0.186	0.66	-2.991	-3.11***	0.116	0.33	0.462	1.92*	0.038	0.12				
Bids	-0.181	-2.18**	-0.118	-1.29	0.003	0.01	-0.141	-1.35	0.340	4.20***	0.300	2.40***				
Activism	0.558	3.46***	0.830	3.09***	0.077	1.88*	0.344	2.73***	-0.168	-1.31	-0.314	-1.69*				
Change in arbitrage capital	-0.242	-1.16**			-0.342	-2.73***	-0.453	-1.01	-0.206	-1.40	0.001	0.01				
Volume			-1.095	-0.39	-6.561	-0.81	-0.868	-0.24			8.909	3.37***				
Fraction category 3 (retail funds)			0.057	0.29	0.387	0.64	0.160	0.66			-0.200	-1.41				
Fraction category 4 (investment advisers)			-0.414	-2.76***	-0.041	-0.10	-0.331	-1.76*			-0.085	-0.47				
Managerial holdings					-0.325	-1.22										
Turnover of BIDDERS (cross effect)									0.405	5.33***			0.424	5.84***		
Year and industry dummies	—	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
Mean squared error	0.24		0.22		0.18		0.16		0.17		0.15		0.13			
Adjusted R ²	0.093		0.140		0.159		0.106		0.162		0.113		0.126			
N	3,814		2,863		2,340		238		1,471		1,800		984			

The symbols ***, **, and * denote significance levels of 1%, 5%, and 10%, respectively, for the two-tailed hypothesis test that the coefficient equals zero.

- The result remains negative and significant when the paper controls for stock liquidity, mutual fund ownership, investment adviser ownership, managerial holdings, and other shareholder variables.
- Using the actual offer premium instead of the abnormal return premium gives the same qualitative result.
- The cross-effect is also important: bidder investor turnover has a positive effect on target premia. This means short-term bidder shareholders are associated with bidders paying more to targets.
- Interpretation: short-term target shareholders reduce the target's ability to hold out, while short-term bidder shareholders allow bidders to overpay.

5.6 Table 4: bidder abnormal return regressions

Table 4
Ordinary least squares estimates of the relation between investor turnover and the bidder's short-term stock price performance
We keep the first bid made by each different acquirer within a given context. The left-hand side variable in Columns 1 through 4 is the bidder's abnormal return, defined as the cumulative abnormal return, measured relative to a CRSP value-weighted market regression using a year of prior daily data, to the bidder firm's stock for trading days $[-63, +126]$ relative to the announcement date (Schwert, 2000). Columns 5 and 6 replicate Columns 1 and 4 but use as a dependent variable the sum of the bidder abnormal returns in the $[-1, +1]$ event window around the announcement date. The right-hand side variables are all measured for the bidder firm following the procedures detailed in the caption to Table 3. T-statistics are calculated using White's heteroskedastic consistent errors.

Dependent variable	Bidder abnormal return								Bidder abnormal return in days $[-1, +1]$							
	(1)				(2)				(3)				(4)			
	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic	Coefficient	T-statistic
Intercept	0.947	0.53	0.426	1.86*	0.744	1.65***	1.099	5.39***	0.784	3.71***	-1.07	-0.014	-0.48			
ROE	-0.007	-0.16	0.041	0.61	-0.056	-1.47*	-0.096	-1.95*	-0.063	-1.85*	0.000	-3.64***	0.000	-0.94		
Sales growth	-0.033	-0.46	0.001	0.05	-0.019	-0.78	0.040	1.09	-0.007	-0.33	0.000	0.01	-0.007	-1.02		
Liquidity	-0.021	0.36	-0.159	-2.36**	-0.144	-2.10**	-0.071	-0.87	-0.159	-2.31**	-0.001	-0.07	-0.003	-0.29		
D/E	-0.001	-0.20	-0.001	-0.10	-0.009	-1.28	0.090	1.53	-0.011	-1.46	0.000	-0.02	0.000	0.31		
M/B	-0.001	-0.51	0.001	0.49	0.000	-0.17	-0.010	-2.34**	0.000	-0.02	0.000	-1.38	0.000	-0.58		
P/E	-0.000	0.64	0.000	0.97	0.000	-0.17	0.000	-0.32	0.000	1.23	0.000	-1.80*	0.000	-1.43		
Size	-0.008	-1.39	-0.030	-3.11***	-0.030	-3.18***	-0.027	-1.97*	-0.030	-3.11***	0.002	1.51	0.002	1.55		
Hostility	-0.055	-1.72	-0.031	-0.97	-0.042	-1.40	-0.048	-1.23	-0.039	-1.29	-0.011	-2.81***	-0.011	-2.61***		
Competing bidders	-0.014	-0.53	0.003	0.09	0.014	0.45	0.047	0.79	0.009	0.27	0.004	-0.30	-0.004	-0.92		
Intra-industry	-0.003	0.16	0.033	1.71*	0.039	2.05**	0.010	0.38	0.038	2.05**	0.000	-0.13	-0.001	-0.27		
Tender offer	-0.042	1.99	0.006	0.26	-0.003	-0.13	0.000	0.26	-0.007	-0.13	0.004	1.07	0.004	1.23		
Threshold	-0.002	0.09	0.027	0.86	0.043	1.51	0.043	0.99	0.038	1.26	0.005	0.91	0.004	0.80		

Investor turnover	-0.519	-2.50**	-0.354	-1.02***	-0.478	-2.51**	-0.452	-2.90***	-0.028	-1.97**	-0.039	-2.15
Fraction	-0.019	-0.35	0.143	1.25	0.122	0.60	0.069	0.74	0.005	0.66	-0.024	-1.51
Concentration	-0.234	-1.25	-0.066	-0.28	-0.517	-1.16**	-0.025	-0.10	0.029	0.38	0.041	1.22
Manager			0.448	0.93	0.490	1.00	1.866	2.78**	0.659	1.45	-0.037	-0.58
Concentration												
Industry exposure	-0.461	-1.18	-0.640	-1.40	-1.967	-3.01***	-8.759	-1.93*	-0.008	-0.17	-0.027	-0.52
Bids	-0.067	-0.48	-0.203	-1.62	-0.266	-1.52	-0.242	-1.90*	-0.019	-1.11	-0.030	-1.61
Activism	-0.569	-3.32***	-0.439	-2.90**	-0.234	-0.65	0.421	2.16**	-0.028	-1.25	-0.017	-0.61
Change in arbitrage capital	-0.119	-1.0	-0.143	-1.28	-0.145	-0.82	-0.352	-1.37	-0.003	-0.23	-0.004	-0.24
Volume	1.830	0.48	14.830	2.85***	1.160	0.81			0.975	2.21**		
Fraction category 3 (retail funds)	-0.074	-0.36	-0.313	-1.05	0.000	0.00			0.040	1.75*		
Fraction category 4 (investment advisers)	-0.424	-2.38**	-0.662	-2.36**	-0.366	-2.01**			0.028	1.14		
Managerial holdings					0.069	0.24						
Turnover of TARGET (cross-effect)					0.064	0.72			0.005	0.40		
Year and industry dummies	—	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean squared error	0.20		0.17		0.15		0.13		0.00		0.00	
Adjusted R ²	0.081		0.034		0.048		0.062		0.011		0.008	
N	2,202		1,964		1,842		704		1,804		1,804	

The symbols ***, **, and * denote significance levels of 1%, 5%, and 10%, respectively, for the two-tailed hypothesis test that the coefficient equals zero.

Read:

- The main coefficient is the coefficient on bidder investor turnover.
- Across specifications, bidder investor turnover has a **negative and statistically significant** effect on bidder abnormal returns.
- This supports the idea that short-term bidder shareholders provide weaker monitoring.
- The effect is weaker in the narrow $[-1, +1]$ announcement window than in the longer Schwert-style abnormal return window.

- Interpretation: the market may take time to learn the true quality and terms of the deal, so the value effect is spread over a broader window.

5.7 Table 5: probability of acquisition and sample-selection correction

Table 5

Full analysis of the relation between investor turnover and the acquisition premium taking into account possible sample-selection issues
Panel A presents a probit regression of the likelihood of a firm being a target (bidder) of a takeover bid. The base sample universe is the merged set of COMPUSTAT, CDA/Spectrum, and CRSP firms and includes a total of 17,640 different securities tracked over each of the sample years 1980 through 1999. The number of observations available refers therefore to firm-years instead of events as in previous tables. The left-hand side variable of the probit regression is a dummy taking the value of one if the company is the target (bidder) of an acquisition bid that year and zero otherwise. Right-hand side variables include all accounting and institutional variables calculated as before (see Table 3), with the exception that they are now calculated as averages over the prior two years. We also add the cumulative abnormal return, measured relative to a CRSP value-weighted market model and estimated using the third year prior to the forecast year, of the firm's stock for the two previous years (see Palepu, 1986; Comment and Schwert, 1995). Panel B presents the estimates for the relation between investor turnover and the acquisition premium using Heckman's sample-selection correction technique (Heckman, 1979). This equation is estimated using the selected sample (i.e., the sample of firms for which there were events). The left-hand side variable is, as before, the target's (bidder's) premium, while target (bidder) accounting and institutional variables are used on the right-hand side. The results for targets are presented in the left-hand side of the table, while results for bidders are presented in the right-hand side. The statistical significance of Heckman's Lambda can also be seen as a test for the null hypothesis that sample selection is irrelevant in the sample. *T*-statistics for the second stage are calculated using the procedure described in Greene (1997).

	Likelihood of being a TARGET firm			Likelihood of being a BIDDER firm		
	Coefficient	T-statistic	Marginal effect	Coefficient	T-statistic	Marginal effect
<i>Panel A. First stage: probit estimates</i>						
Intercept	-2.63	-23.10	-0.18***	-1.87	-17.82	-0.18***
ROE	0.00	-0.46	0.00	0.00	1.92	0.00
Sales growth	0.02	0.85	0.00	0.15	7.87	0.02**
Liquidity	-0.10	-1.79	-0.01*	-0.28	-5.54	-0.03***
<i>D/E</i>	0.01	2.01	0.00*	0.00	1.46	0.00
<i>M/B</i>	-0.01	-1.83	0.00*	0.00	1.11	0.00
<i>P/E</i>	0.00	-1.19	0.00	0.00	1.08	0.00
Size	0.00	-5.56	0.00***	0.00	18.32	0.00***
Abnormal return	-61.92	-6.76	-4.18***	28.64	3.66	2.78***
Investor turnover	1.29	6.83	0.09***	0.94	5.23	0.09***
Fraction	0.00	6.19	0.03***	0.61	11.58	0.06***
Concentration	-0.99	-7.42	-0.07***	-1.67	-13.35	-0.16***
Manager concentration	0.50	0.86	0.72	1.40	0.07	0.00
Industry exposure	0.50	1.63	0.03	0.56	2.07	0.05**
Beta	0.32	3.03	0.02***	-0.16	-1.65	-0.02*
Activism	-0.10	-0.54	-0.01	0.62	3.72	0.06***
Change in arbitrage capital	1.58	4.75	0.11***	-0.23	-0.84	-0.02
Likelihood ratio index		0.39			0.10	
<i>N</i>		43,793			44,005	

Dependent variable	TARGET abnormal return premium		TARGET actual offer premium		BIDDER abnormal return premium	
	Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic
<i>Panel B. Sample-selection corrected estimates for premium regressions</i>						
Intercept	2.40	8.90***	0.67	2.47**	0.72	2.88***
ROE	0.00	0.34	-0.01	-0.74	-0.07	-2.03**
Sales growth	-0.04	-1.28	-0.04	-1.13	-0.02	-1.16
Liquidity	0.01	0.23	-0.12	-2.03**	-0.04	-0.74
<i>D/E</i>	-0.02	-0.34	0.03	0.46	-0.02	-3.06***
<i>M/B</i>	-0.01	-1.94*	0.01	1.36	0.00	0.30
<i>P/E</i>	0.00	-0.56	0.00	-0.64	0.00	0.53
Size	-0.06	-5.49***	-0.04	-3.64***	-0.03	-2.38**
Hostility	0.03	0.68	0.02	0.52	-0.04	-1.27
Competing bidders	0.04	0.83	-0.01	-0.13	0.00	-0.01
Intra-industry	-0.02	-0.93	0.03	1.22	0.03	1.62
Tender offer	0.19	3.85***	0.05	1.77*	0.01	0.24
Toehold	-0.10	-3.18***	-0.11	-3.31***	0.72	2.84***
Target investor turnover	-0.55	-5.13***	-0.25	-2.24**	0.20	1.90**
Bidder investor turnover	0.49	6.76***	0.47	6.22***	-0.32	-2.63**
Fraction	0.07	0.48	0.24	1.62	0.11	0.92
Concentration	0.54	2.80***	0.37	1.98**	0.15	0.81
Manager concentration	-0.49	-0.97	-0.58	-0.91	0.53	1.07
Industry exposure	0.04	0.13	0.20	0.66	-0.79	-3.08***
Beta	-0.30	-3.23***	0.15	1.65*	-0.22	-2.36**
Activism	0.57	2.83***	-0.29	-1.46	0.40	2.18**
Change in arbitrage capital	0.03	0.20	0.10	0.78	0.02	0.22
Volume	-1.06	-0.55	10.63	4.16*	4.07	1.56
Fraction category 3 (m. funds)	0.29	1.36	-0.13	-0.61	-0.07	-0.42
Fraction category 4 (inv. Adv.)	-0.26	-1.41	-0.02	-0.12	-0.49	-2.96***
Heckman's lambda	-0.49	-3.35***	-0.01	-0.10	-0.04	-0.72
Mean squared error		0.17		0.13		0.15
Adjusted <i>R</i> ²		0.170		0.137		0.048
<i>N</i>		1,354		969		1,702

The symbols ***, **, and * denote significance levels of 1%, 5%, and 10%, respectively, for the two-tailed hypothesis test that the coefficient equals zero.

Read:

- Panel A estimates probit models for the likelihood of becoming a target or bidder.
- Investor turnover is positive and significant for both target likelihood and bidder likelihood.
- The marginal effect of investor turnover is about **0.09** for both being a target and being a bidder.
- Panel B applies Heckman's correction to the premium regressions.
- The target investor-turnover coefficient remains negative, while the bidder investor-turnover cross-effect remains positive.
- The bidder return result also survives the selection correction.
- Interpretation: short-term shareholders affect both the probability and the price of takeover activity, but the premium and return results are not just sample-selection artifacts.

Equally weighted portfolios						Value weighted portfolios						
(1) After one year		(2) After two years		(3) After three years		(4) After one year		(5) After two years		(6) After three years		
Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic	Coefficient	<i>T</i> -statistic	
<i>Panel A. Portfolio of all acquirers</i>												
α	-0.001	-1.48	-0.004	-4.19***	-0.004	-4.44***	0.001	1.11	-0.002	-2.55**	-0.002	-2.40**
$\beta_{\text{turn-RE}}$	1.078	46.80***	1.082	46.19***	1.096	42.43***	1.042	34.10***	1.047	43.72***	1.045	39.16***
β_{turn}	0.630	13.87***	0.688	17.64***	0.710	17.13***	-0.171	-3.50***	-0.075	-1.90**	-0.057	-1.41
β_{turnL}	0.115	2.28**	0.164	3.82***	0.207	5.02***	-0.083	-1.45	-0.002	-0.04	0.063	1.27
Adjusted <i>R</i> ²	0.93		0.95		0.95		0.88		0.92		0.90	
<i>N</i>	190		190		190		190		190		190	
SR	-0.073	-1.05	-0.204	-2.92***	-0.176	-2.15**	0.022	0.32	-0.183	-2.62**	-0.150	-2.15**
<i>Panel B. Portfolio of acquirers with high investor turnover minus acquirers with low investor turnover</i>												
$\alpha_{\text{HT minus LT}}$	-0.005	-2.40**	-0.005	-2.63***	-0.006	-3.09***	-0.002	-0.80	-0.007	-2.74***	-0.007	-2.64***
$\beta_{\text{turn-RE}}$	0.185	3.05***	0.172	3.03***	0.187	3.49***	0.194	2.65***	0.221	2.89***	0.275	4.08***
β_{turn}	0.328	3.96***	0.341	4.19***	0.380	5.53***	0.544	4.33***	0.631	6.54***	0.528	5.54***
β_{turnL}	-0.255	-2.23**	-0.322	-3.20***	-0.306	-3.36***	-0.222	-1.42	-0.157	-1.16	-0.164	-1.67*
Adjusted <i>R</i> ²	0.20		0.29		0.36		0.18		0.27		0.31	
<i>N</i>	190		190		190		190		190		190	
SR _{HT minus LT}	-0.118	-1.68*	-0.215	-3.04***	-0.254	-3.59***	-0.045	-0.063	-0.141	-1.99**	-0.139	-1.97**
<i>Panel C. Portfolio of only acquirers with high investor turnover</i>												
α_{HT}	-0.005	-3.64***	-0.007	-4.90***	-0.007	-5.14***	-0.002	-0.90	-0.218	-3.11***	-0.007	-4.16***
SR _{HT}	-0.146	-2.07**	-0.284	-4.04***	-0.308	-4.38***	-0.085	-1.21	-0.168	-2.40**	-0.229	-3.26***
<i>Panel D. Portfolio of only acquirers with low investor turnover</i>												
α_{LT}	0.000	0.25	-0.002	-1.30	-0.002	-1.32	0.001	0.25	0.000	0.20	0.000	-0.15
SR _{LT}	-0.001	-0.02	-0.064	-0.90	-0.050	-0.71	-0.059	-0.83	-0.044	-0.62	-0.073	-1.03

The symbols ***, **, and * denote significance levels of 1%, 5%, and 10%, respectively, for the two-tailed hypothesis test that the coefficient equals zero.

5.8 Table 6: long-run acquirer performance

Read:

- The paper forms calendar-time portfolios of acquirers after completed acquisitions.
- Panel A shows that acquirers as a group underperform over longer horizons, consistent with prior M&A literature.
- Panel B buys acquirers with high investor turnover and sells acquirers with low investor turnover.
- The high-minus-low strategy has a negative abnormal return. Over a three-year holding period, the value-weighted alpha is about **−0.007 per month**.
- This is economically large: roughly **−8% per year**.
- Panels C and D show that the effect is mostly driven by underperformance of high-turnover acquirers, not strong positive performance of low-turnover acquirers.
- Interpretation: weak monitoring by short-term shareholders does not only affect announcement returns; it is also linked to persistent post-merger underperformance.

5.9 Additional robustness checks

Read:

- The paper reports that the main results survive excluding merger arbitrageurs from the investor-turnover measure.
- Results are robust to dropping other shareholder portfolio controls.
- Results are similar when splitting the sample by acquisition mode, such as mergers versus tender offers.
- For bidders, the negative effect of investor turnover is partially mitigated when managerial incentives are more aligned with shareholders through equity-based compensation.
- This supports the monitoring interpretation: if managers are already better aligned, weak shareholder monitoring matters less.

6 Short summary

- **Method.** Construct a firm-level measure of shareholder investment horizon from institutional investors' portfolio churn rates.
- **Identification logic.** Measure shareholder horizon well before the M&A event; control for ownership level, concentration, investor type, liquidity, activism, arbitrage capital, and deal characteristics; then check bid likelihood, premium, bidder return, selection, and long-run performance.
- **Target result.** Short-term target shareholders make bids more likely but reduce acquisition premia. A one-standard-deviation increase in turnover reduces the target premium by about **3 percentage points**.

- **Bidder result.** Short-term bidder shareholders are associated with worse bidder announcement returns and worse long-run post-merger returns.
- **Cross-effect.** Short-term bidder shareholders raise target premia, consistent with bidder overpayment.
- **Main conclusion.** Shareholder horizons matter because they affect both the bargaining power of targets and the monitoring of bidder managers.

7 Conclusion

7.1 Core conclusion

- The paper shows that institutional shareholders' investment horizons are an important determinant of takeover outcomes.
- Target firms with short-term shareholders are easier and cheaper to acquire.
- Bidder firms with short-term shareholders are more likely to make value-reducing acquisitions.
- The market for corporate control therefore depends not only on ownership concentration or institutional ownership levels, but also on how long shareholders are willing to stay invested.

7.2 Economic intuition

- Long-term shareholders can credibly wait for the full value of a takeover to be realized, so they strengthen the target's bargaining position.
- Long-term shareholders also have stronger incentives to monitor managers, because they expect to remain shareholders long enough to capture the benefits of monitoring.
- Short-term shareholders are more likely to sell than to monitor or hold out.
- This makes targets more affordable and gives bidder managers more room to overpay or pursue private benefits.

7.3 Takeaway

The key lesson is that shareholder identity matters through investment horizon. Short-term investors can increase takeover activity by lowering resistance and making deals easier to complete, but this comes with a governance cost: lower target premia and more value-destroying bidding. Long-term capital is scarce and valuable because it supports both stronger bargaining and stronger managerial discipline.